

1.

```
package com.payoda.training1.ooc;
interface LibraryUser {
    void registerAccount();
    void requestBook();
}
class KidUser implements LibraryUser {
    int age;
    String bookType;
    public KidUser(int age, String bookType) {
        this.age = age;
        this.bookType = bookType;
    }
    public void registerAccount() {
        if (age < 12) {
            System.out.println("You have successfully registered under a Kids Account");
        } else {
            System.out.println("Sorry, Age must be less than 12 to register as a kid");
        }
    }
    public void requestBook() {
        if (bookType.equalsIgnoreCase("Kids")) {
            System.out.println("Book Issued successfully, please return the book within 10 days");
        } else {
            System.out.println("You are allowed to take only kids books");
        }
    }
}
class AdultUser implements LibraryUser {
    int age;
    String bookType;
    public AdultUser(int age, String bookType) {
        this.age = age;
        this.bookType = bookType;
    }
    public void registerAccount() {
        if (age > 12) {
            System.out.println("You have successfully registered under an Adult Account");
        } else {
            System.out.println("Sorry, Age must be greater than 12 to register as an adult");
        }
    }
    public void requestBook() {
        if (bookType.equalsIgnoreCase("Fiction")) {
            System.out.println("Book Issued successfully, please return the book within 7 days");
        } else {
            System.out.println("You are allowed to take only adult Fiction books");
        }
    }
}
```

```

    }
}
}
public class LibrarySystem {
    public static void main(String[] args) {
        LibraryUser kid1 = new KidUser(10, "Kids");
        kid1.registerAccount();
        kid1.requestBook();

        LibraryUser kid2 = new KidUser(13, "Fiction");
        kid2.registerAccount();
        kid2.requestBook();
        LibraryUser adult1 = new AdultUser(25, "Fiction");
        adult1.registerAccount();
        adult1.requestBook();
        LibraryUser adult2 = new AdultUser(10, "Kids");
        adult2.registerAccount();
        adult2.requestBook();
    }
}

```

Output :

```

You have successfully registered under a Kids Account
Book Issued successfully, please return the book within 10 days
Sorry, Age must be less than 12 to register as a kid
You are allowed to take only kids books
You have successfully registered under an Adult Account
Book Issued successfully, please return the book within 7 days
Sorry, Age must be greater than 12 to register as an adult
You are allowed to take only adult Fiction books

```

2.

```

package com.payoda.training1.ooc;
import java.util.*;
public class MergeSortArrayList {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ArrayList<Integer> list1 = new ArrayList<>();
        ArrayList<Integer> list2 = new ArrayList<>();
        System.out.println("Enter 5 integers for first list:");
        for (int i = 0; i < 5; i++) {
            list1.add(sc.nextInt());
        }
        System.out.println("Enter 5 integers for second list:");
    }
}

```

```

    for (int i = 0; i < 5; i++) {
        list2.add(sc.nextInt());
    }
    ArrayList<Integer> mergedList = new ArrayList<>();
    mergedList.addAll(list1);
    mergedList.addAll(list2);
    Collections.sort(mergedList);
    System.out.println("Merged and Sorted List: " + mergedList);
    ArrayList<Integer> result = new ArrayList<>();
    int[] indices = {2, 6, 8};
    for (int index : indices) {
        if (index < mergedList.size()) {
            result.add(mergedList.get(index));
        }
    }
    System.out.println("Extracted Elements: " + result);
}
}

```

Output:

```

Enter 5 integers for first list:
10
32
34
213
345
Enter 5 integers for second list:
23
64
78
93
23
Merged and Sorted List: [10, 23, 23, 32, 34, 64, 78, 93, 213, 345]
Extracted Elements: [23, 78, 213]

```

3.

```

package com.payoda.training1.ooc;
import java.util.*;
public class StudentGradeChecker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        HashMap<String, Float> studentMap = new HashMap<>();
        System.out.print("Enter number of students: ");
        int n = sc.nextInt();
        sc.nextLine();
        for (int i = 0; i < n; i++) {

```

```

        System.out.print("Enter student name: ");
        String name = sc.nextLine();
        System.out.print("Enter student mark: ");
        float mark = sc.nextFloat();
        sc.nextLine();
        studentMap.put(name, mark);
    }
    System.out.print("Enter student name to check grade: ");
    String query = sc.nextLine();
    if (studentMap.containsKey(query)) {
        float mark = studentMap.get(query);
        if (mark >= 60) {
            System.out.println("Grade for " + query + ": PASS");
        } else {
            System.out.println("Grade for " + query + ": FAIL");
        }
    } else {
        System.out.println("Student not found.");
    }
}
}

```

Output:

```

Enter number of students: 4
Enter student name: dharanesh
Enter student mark: 32
Enter student name: karthi
Enter student mark: 85
Enter student name: surya
Enter student mark: 85
Enter student name: sanjeev
Enter student mark: 99
Enter student name to check grade: sanjeev
Grade for sanjeev: PASS

```

4.

```

package com.payoda.training1.ooc;
import java.util.*;
public class EvenOddSplitter {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ArrayList<Integer> inputList = new ArrayList<>();
        ArrayList<Integer> evenList = new ArrayList<>();
        ArrayList<Integer> oddList = new ArrayList<>();
        System.out.print("Enter number of integers: ");
        int n = sc.nextInt();
    }
}

```

```

System.out.println("Enter " + n + " integers:");
for (int i = 0; i < n; i++) {
    int num = sc.nextInt();
    inputList.add(num);
    if (num % 2 == 0) {
        evenList.add(num);
    } else {
        oddList.add(num);
    }
}
System.out.println("Original List: " + inputList);
System.out.println("Even Numbers: " + evenList);
System.out.println("Odd Numbers: " + oddList);
}
}

```

Output:

```

Enter number of integers: 10
Enter 10 integers:
23
35
56
78
932
73
82
62
39
89
Original List: [23, 35, 56, 78, 932, 73, 82, 62, 39, 89]
Even Numbers: [56, 78, 932, 82, 62]
Odd Numbers: [23, 35, 73, 39, 89]

```